

Collaborative based + Popular based

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Popularity based recommendation system.

Criteria $\rightarrow$ 250 rating minimum.

① merge

② rating-with-name = ratings.merge(book, on='ISBN')

rating-with-name.groupby('Book-Title').count

↓
pandas group all series having the same book title

num-rating-df = ratings-with-name.groupby('Book-Title').count()['Book-rating'].reset_index()

How to rename column.

num-rating-df = num-rating-df.rename(columns={'Book-rating': 'num-ratings'}, inplace=True)

num-ratings-df

→ for the rename permanent

count() $\rightarrow$ how many rated books are there

Series →

Databse

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reset-index as -

Put those row labels into a normal col

Without reset-index:-

Book-Title

~~Book-Rating~~

Harry Potter

3
2

Atomic Habits

With

Book-Title

Book Rating

Harry Potter

3
2

Atomic Habits

Making another columns

→ Book Title

Avg-rating
mean

because when we are making a
popular based system we need user
interaction especially on votes

so that IMDB and Rotten Tomatoes suggest
top 100 movies with specific
formula.

currently we are using more votes
books with atleast 250 votes on her


```

avg-rating-df = C
rating-with-name // df
- group by ('Book-Title') ['Book-Rating']
  • mean()
  • reset_index()

```

groupby('Book-Title') → groups together all rows having the same book-title

['Book-Rating'] → From every group, I only care about the Book-Rating column.

Without this, Pandas would also try to calculate means of Author - Publisher - Genre left.

merging avg-rating-df with ~~num-rating-df~~ num-rating-df.

```

popular-df = num-rating-df.merge(
    pop avg-rating-df, on='Book-Title'
)
popular-df

```

base :-

```

popular-df = popular-df[popular-df
    ['num-ratings'] >= 250]

```


sort with avg rating because
250 db Benchmark already completed

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```
popular_df[popular_df['num-ratings'] > 250]
    .sort_values(['avg-rating', ascending=False])
```

→ giving only num of rated books which are rated by 250+ people.

→ sort values is there to sort the average rating with that more than 250 people has voted with mean and average in desc order.

Now going on production-deployment I need

author name

image

published

already in Database

```
popular_df = (popular_df
```

```
    .merge(books, on='Book-Title')
```

```
    .drop_duplicates(['Book-Title'])
```

```
    [['Book-Title', 'Book-Author', 'Publisher',  
     'Image-URL-M', 'num-ratings',  
     'avg-ratings']])
```

→ Harry Potter have many books so we use this

Renaming the df

avg-rating-df.rename(columns = {'Book rating' :
'avg-rating'})

In place = 'True' }